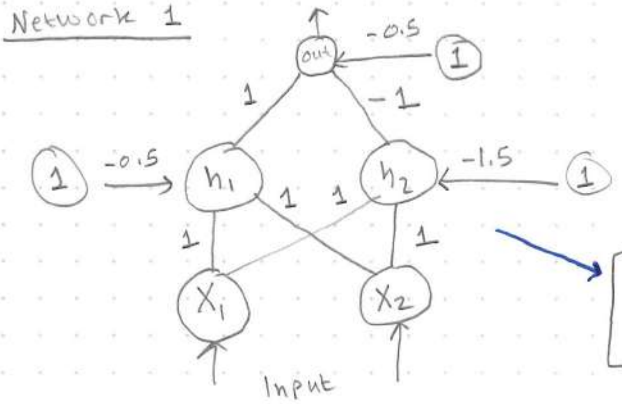
DATE / NO. 2/4/26

TITLE Homework Assignment #1

Show if the following two networks Solve the XOR problem by constructing their a.) decision regions, and b.) their truth tables (need h_1 & h_2)

Network 1



Input	Output
0 0	0
0 1	1
1 0	1
1 1	0

$h_1 = \text{Step}(X_1 + X_2 - 0.5)$

$h_2 = \text{Step}(X_1 + X_2 - 1.5)$

$\text{out} = \text{Step}(h_1 - h_2 - 0.5)$

Decision Regions

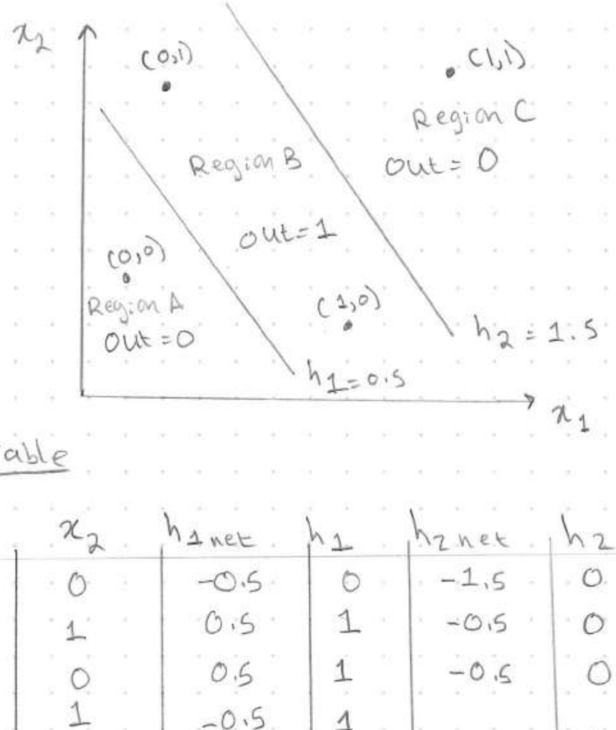
h_1 boundary : $X_1 + X_2 = 0.5 \longrightarrow h_1 = 1$ when $x_1 + x_2 \geq 0.5$

h_2 boundary : $X_1 + X_2 = 1.5 \longrightarrow h_2 = 1$ when $x_1 + x_2 \geq 1.5$

Region A : $(X_1 + X_2 < 0.5)$: $h_1 = 0$, $h_2 = 0 \longrightarrow \text{Out} = \text{Step}(-0.5) = \underline{0}$ [Contains (0,0)]

Region B : $(0.5 \leq x_1 + x_2 < 1.5)$: $h_1 = 1$, $h_2 = 0 \longrightarrow \text{Out} = \text{Step}(0.5) = \underline{1}$ [Contains (0,1), (1,0)]

Region C : $(x_1 + x_2 \geq 1.5)$: $h_1 = 1$, $h_2 = 1 \longrightarrow \text{Out} = \text{Step}(-0.5) = \underline{0}$ [Contains (1,1)]

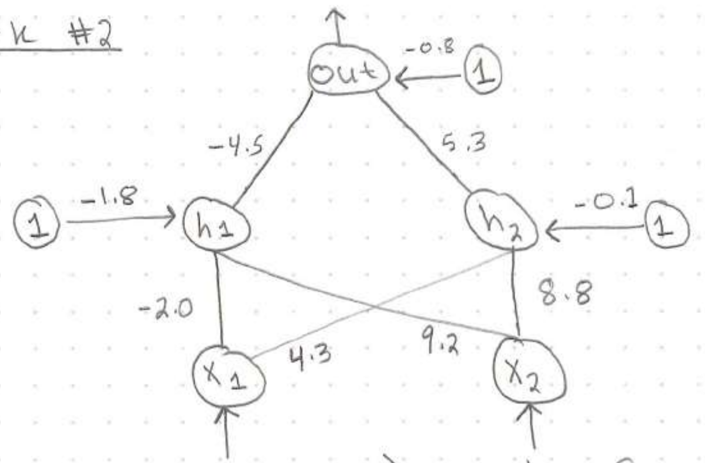


Truth Table

x_1	x_2	h_1 net	h_1	h_2 net	h_2	out net	out	XOR	
0	0	-0.5	0	-1.5	0	-0.5	0	0	✓
0	1	0.5	1	-0.5	0	0.5	1	1	✓
1	0	0.5	1	-0.5	0	0.5	1	1	✓
1	1	-0.5	1	0.5	1	-0.5	0	0	✓

Network #1 Solves the XOR Problem

Network #2



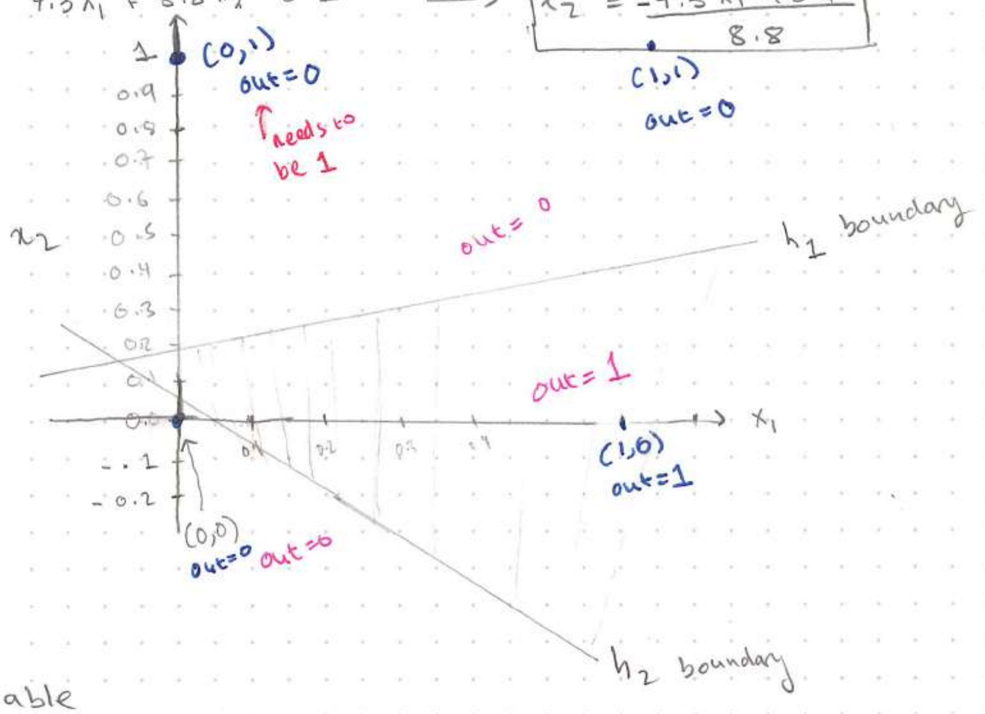
Input	Output
0 0	0
0 1	1
1 0	1
1 1	0

- $h_1 = Step(-2.0x_1 + 9.2x_2 - 1.8)$
- $h_2 = Step(4.3x_1 + 8.8x_2 - 0.1)$
- $Out = Step(-4.5h_1 + 5.3h_2 - 0.8)$

Decision Regions

$$h_1 = -2x_1 + 9.2x_2 - 1.8 = 0 \longrightarrow \boxed{x_2 = \frac{2x_1 + 1.8}{9.2}}$$

$$h_2 = 4.3x_1 + 8.8x_2 - 0.1 = 0 \longrightarrow \boxed{x_2 = \frac{-4.3x_1 + 0.1}{8.8}}$$



Truth Table

x_1	x_2	h_1 net	h_1	h_2 net	h_2	Out net	Out	XOR	
0	0	-1.8	0	-0.1	0	-0.8	0	0	✓
0	1	7.4	1	8.7	1	0.0	1	1	✓
1	0	-3.8	0	4.2	1	4.5	1	1	✓
1	1	5.4	1	13.0	1	0	1	0	X

Network #2 does <u>not</u> solve the XOR problem. The hidden layer fails to create distinct representations for (0,1) and (1,1), making it impossible for the output layer to produce different results for these inputs